

Tips for interacting with partitioned tables

So far, you've learned that a partitioned table is a large table that's divided into smaller segments or partitions. One of the primary reasons for partitioning is to have smaller subsets of data to work with and optimize query performance. When working with partitioned tables, you can manage, add, and export data in individual partitions. In this reading, you'll learn how to manage partitioned data in BigQuery.

Manage partitioned data

Once data is partitioned, your primary focus may be querying less data and creating reports using more focused data to improve performance. Using table partitioning allows you to set and control partition expiration dates, whether partition filters are required in your queries, and add, copy, export, or remove individual partitions without affecting the rest of the table.

Loading data

You can load data to a specific partition by using the `bq load` command with a partition decorator. When writing the command, specify the partition and the file you want to upload. For example, if you have a file called `mydataset.csv`, and you want to add it to a specific partition in the `mydata.subset` table, you'd write `mydata.subset$v2` where *v2* is the partition decorator. The full command is:

```
bq load --source_format=CSV 'mydata.subset$v2' mydataset.csv
```

For example, the day partition corresponding to May 1, 2016 could be referenced as `'mydata.subset$20160501'`.

For tables partitioned on a time-unit or integer column, data will be sorted into the correct partitions automatically, even without specifying the partition decorator.

Exporting data

To export data from an individual partition, use the `extract` command written `bq extract`. To use this command for a partition, add the partition decorator to the table name. For example, if you're working with the table `'mydata.subset'`, you could add a partition decorator for the day partition May 1, 2019. You would reference this partition as `'mydata.subset$20190501'`.

Key takeaways

Once your data is partitioned, you can manage the table just like any other BigQuery table. You can also manage individual partitions without affecting the rest of the table, like adding,

removing, or exporting partitions and data. Partitioning also lets you set additional policies like partition expiration and partition filter requirements.

Resources for more information

Google Cloud documentation provides detailed information about managing and editing datasets. To learn more, check out these links:

- Google provides advanced queries for managing partitioned tables:
<https://cloud.google.com/bigquery/docs/managing-partitioned-tables>
- Google's documentation on loading data into partitioned tables gives additional information about importing data into a partitioned table:
<https://cloud.google.com/bigquery/docs/load-data-partitioned-tables#write-to-partition>